Visualization, Analysis, and Prediction of Parallax Bias in Gaia Data Release 3 Using Data Science Techniques

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Introduction

Stellar distance is commonly measured using a geometric method call parallax, which involves three points forming a triangle: a target object (C) and two observing locations (A and B) that are some baseline apart and at an equal distance from C (Fraknoi et al., 2022). Parallax is the angle subtended by the baseline between A and B (Fraknoi et al., 2022). For astronomy, those two locations are opposite sides of Earth's orbit. Distance can be computed using

$$d = 1/p$$

where d is distance in parsecs and p is the parallax angle.

The Gaia mission provides extremely precise parallaxes for nearly two billion stars but even so, every measurement includes an associated uncertainty. These uncertainties propagate into distance estimates and can introduce systematic biases that affect downstream analyses.

This study investigates how parallax uncertainties in Gaia Data Release 3 influence derived distances and how these biases vary across different stellar environments. The study evaluates how well Gaia attributes can predict parallax error, characterizes where distance estimates remain reliable, and identifies regions or parameter ranges where systematic distortions emerge.

Methodology

The study analyzes four stellar groups, all retrieved from the Gaia archive (GAIA Archive, n.d.), using ADQL: nearby stars as a control sample, Galactic Plane stars as a stress test for crowding and noise, and the Pleiades and Hyades clusters where true distances are well known. The workflow begins with exploratory data analysis to compare the structure and variability of each dataset and to identify high-level relationships. Naïve distance distributions (computed directly from parallax) are examined to reveal how measurement uncertainty propagates into distance estimates. The resulting distance estimates are used to generate stellar classification diagrams. Residual analysis highlights where simple inverse-parallax distances diverge from expectations.

Principal component analysis (PCA) using SIMCA 18.2 is used to characterize dominant variance directions. Two predictive models – LASSO regression and Gradient Boosting – are applied to quantify which attributes most strongly influence parallax error and how predictive structure differs. Finally, Hertzsprung-Russell diagrams are used to visualize how distance biases distort scientific interpretation

Results

This work has successfully shown the impact of measurement uncertainty on downstream analyses and identified attributes with high predictive potential.

The primary method of determining whether measurement uncertainty impacts results is with naïve calculations. Naïve distance calculations (i.e. distance calculations that do not account for uncertainty) are used to calculate the absolute magnitude of stars. Apparent magnitude is a measured value included in our dataset, which represents how bright a star appears from Earth. Absolute magnitude is calculated using distance and represents a star's intrinsic brightness from a set distance of 10 parsecs. Calculating distance from parallax, and absolute magnitude from distance, is a standard practice in astrophysics. Completing these calculations on data with varying amounts of relative error highlights the effect of measurement error on results.

Relative parallax error is calculated as parallax error divided by the parallax value. It is a metric that represents the quality and reliability of parallax measurements. The subset of data nearby stars (stars within 200 parsecs of Earth) has a median relative error of 0.0068. This is significantly lower than the subset of data containing galactic plane stars (stars that exist along the galactic plane, a noisy region with faint stars and observational hazards), which has a median relative error of 0.168, almost two orders of magnitude greater. This difference makes these two subsets ideal comparisons for highlighting the impact of parallax error on results, visualizations, and interpretability.

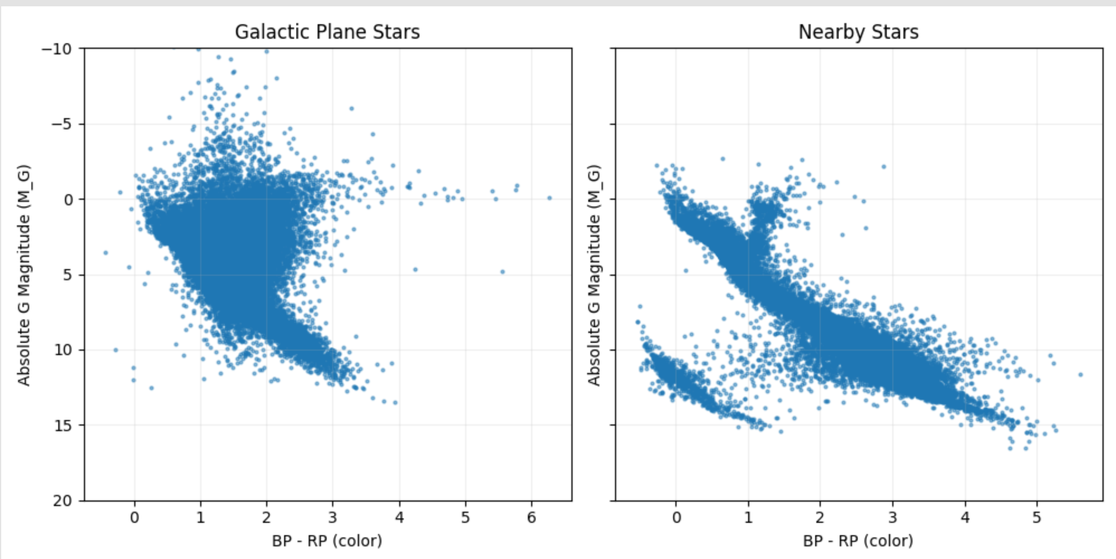

Figure 1: H-R diagrams of galactic plane stars and nearby stars

Hertzsprung-Russel (H-R) diagrams are scatter plots of stars that show the relationship between stars' absolute magnitudes and their color. These diagrams are used to classify stars into categories such as main sequence, giants, and white dwarfs. Figure 1 shows the H-R diagrams created using the subsets of data in this project, with absolute magnitude values calculated with naïve distances.

Comparing the two diagrams, there is a clear distinction. The diagram for nearby stars (right) has a well-defined structure, and clearly identifiable sections for main sequence stars, giants, and white dwarfs. In contrast, the diagram for Galactic Plane stars (left) has a less defined structure and no clear separation between stellar categories.

This distinction is also visible in Figure 2, which plots H-R diagrams using only binned data from the Galactic Plane subset of data. In the diagram created from low-uncertainty data (left), there is clear structure and separation of categories, while the diagram created from high-uncertainty data (right) lacks visible structure or definition.

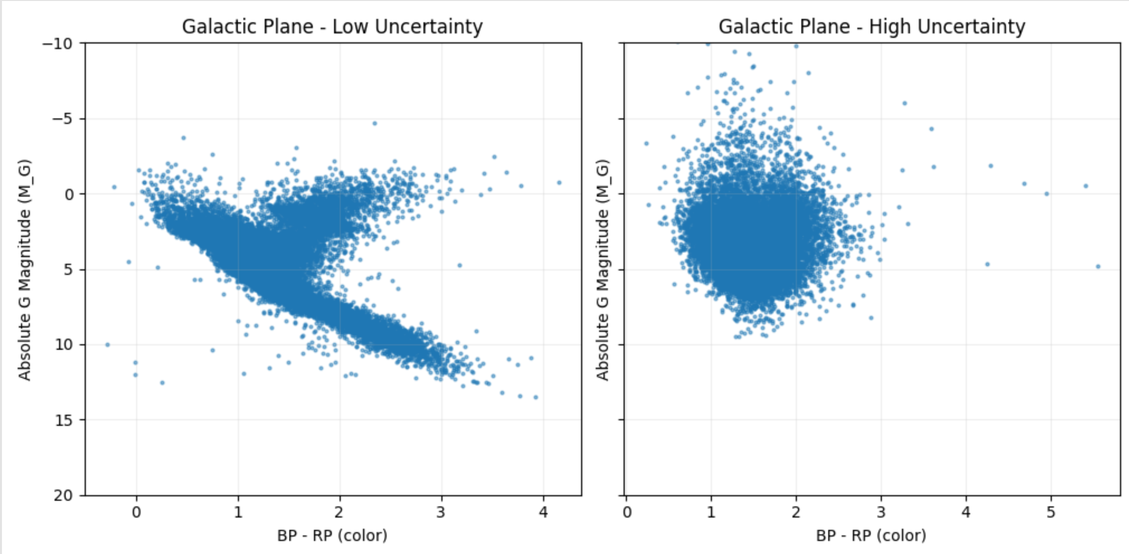

Figure 2: H-R diagrams of galactic plane stars binned by relative uncertainty

Across both LASSO and Gradient Boosting models, parallax error predictors are dominated by indicators of astrometric quality, with excess noise from the source emerging as the strongest feature in most star groups. Nearby stars, Galactic Plane, and Hyades samples also rely on brightness and visibility period, while the Pleiades showed sensitivity to BP-RP colour as well.

The magnitude of feature importances in the Gradient Boosting model is far greater than those of the LASSO regression model, indicating that the structure of the data exhibits complexities that the simpler LASSO model fails to capture. However, because the pattern of strong coefficients for excess noise and G-band mean magnitude is consistent across both models, the LASSO model supports the interpretation of these attributes being the strongest indicators of parallax error – even under linear assumptions.

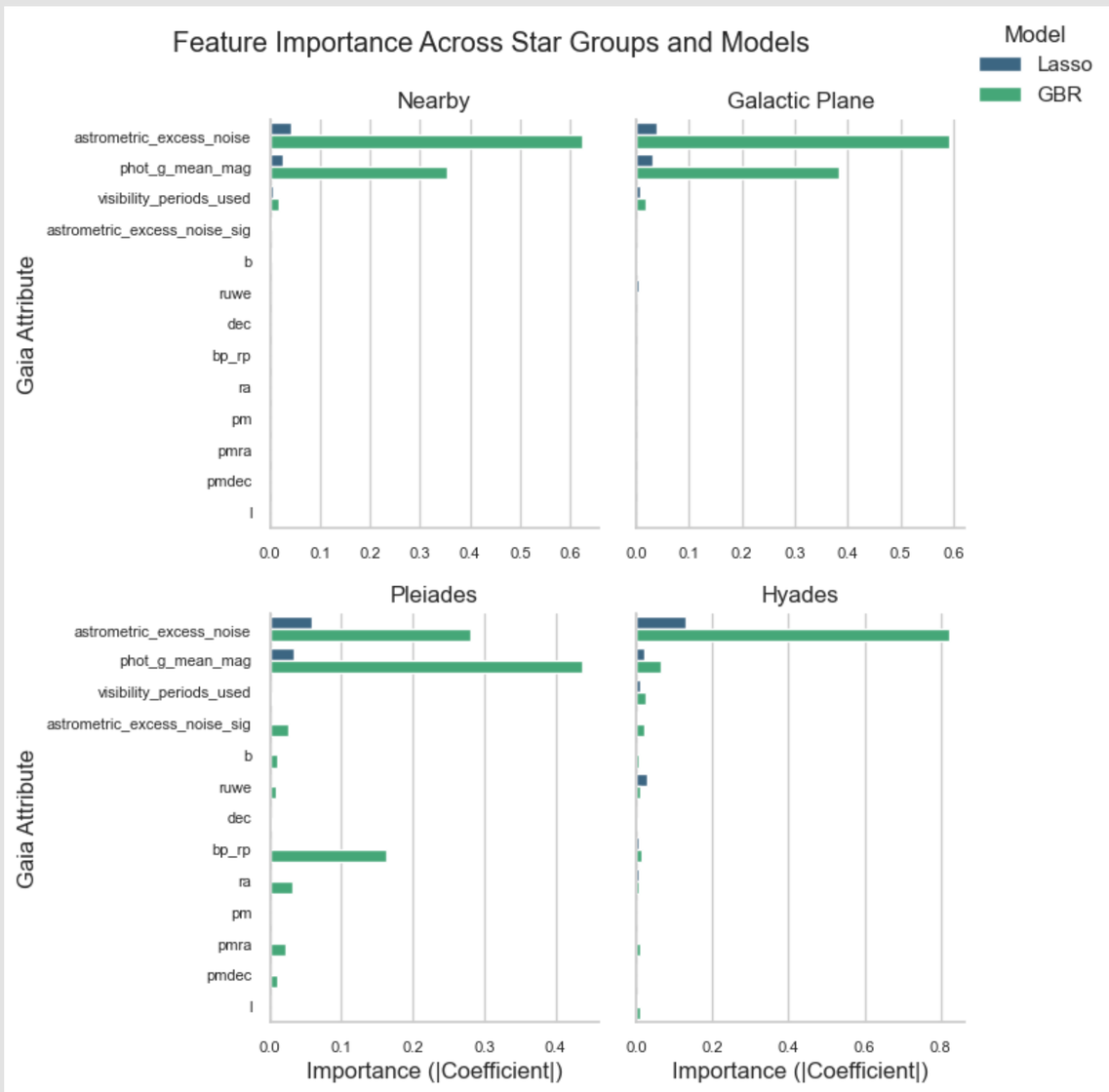

Figure 3: Feature importance for each star group and model

Conclusion

This project demonstrates how measurement uncertainty in the Gaia DR3 dataset has a direct and observable impact on derived quantities and visual analyses. Using naïve distance calculations, the results show that higher relative parallax error leads to increased variability in distance estimates and a clear degradation in H-R diagram structure. These findings support the original hypothesis that uncertainty is not random noise, but a structured, systematic component of data that can significantly affect downstream results.

The comparison between nearby stars and Galactic Plane stars highlights how differences in measurement quality translate into differences in interpretability. And while measurement quality indicators – like excess noise from the source and G-band mean magnitude – drive parallax error prediction most strongly, different stellar populations may express their own structure depending on environmental and intrinsic astrophysical properties.

These results are relevant beyond astronomy, as they emphasize a broader principle: ignoring measurement uncertainty may lead to distorted and misleading conclusions, especially when applying nonlinear transformations. Accounting for uncertainty when calculating values using measured data is crucial for producing accurate, reliable, and interpretable conclusions in any data driven field.

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